

Project Proposal

Start Assignment

- Due Friday by 11:59pm
- Points 3
- Submitting a file upload

Overview

Please create a group on Canvas by going to People -> Final Project Groups and having everyone add themselves to the same group. Group grades and comments will only show up for people who are in the same group on Canvas.

The goals of this first assignment are:

1. To choose a business problem that your team wants to address during the remainder of the course using Machine Learning (ML).
2. To find at least 1 source (from a textbook, from a blog post, ...) where this problem has already been solved using ML.
3. To find at least 1 relevant dataset that will allow your team to build a Proof Of Concept (POC) ML system to address the business problem of your choice over the next few weeks. There is a good chance that relevant datasets are already mentioned in the sources that your team will find as part of the work associated with the previous goal.

In one sentence: by the end of this first assignment, your team needs to be clear on what it will be working on for the next few weeks, and it needs to be confident that the chosen business problem has some existing baseline solution.

The best case scenario is that by the end of the project your team will have implemented a different or improved solution. But your team also needs to mitigate the risk of the worst case scenario - i.e., not being able to deliver any reasonably decent POC ML system by the end of the course - as early as possible. This is precisely why your team will find sources that already addressed the chosen business problem as part of this assignment.

Deliverable

Your team will submit a project proposal of no more than 3 pages describing:

1. The business problem that your team will focus on.
2. How this problem has already been solved in the past using ML (from the sources you found).
3. The datasets that your team has identified.

Your team is welcome to include screenshots and other figures as needed. These do not count towards the 3 page limit.

Your team can submit the project proposal as a PDF document or as a Word document.

Checklist

Here is a checklist to help your team make sure that the most important items are covered in the project proposal.

- ☐ Include a description of the business problem that your team wants to address and why it is an interesting problem to solve from a business perspective. Does solving this problem help making money? Saving money? Saving time? For instance: Given a collection of text documents, we want to help users identify the most frequent topics represented in the collection. This is useful in a number of practical settings. For example: what are the characteristics of a product that are most frequently discussed by its users in their reviews of that product? Developing a better understanding of the perception and experiences that the users have with the product can help a company make money by introducing new features for which the users explicitly indicate interest, and it can save time and money because it provides immediately actionable user feedback without the need of conducting more extensive and more expensive user research.
- ☐ Include at least 1 source where this business problem has been solved using ML. For instance: This type of problem - generally referred to as "topic modeling" - has been addressed in the past in a few ways. For example, SOURCE uses a Latent Dirichlet Allocation (LDA) model to learn the topics represented in COLLECTION OF DOCUMENTS. Similarly, OTHER SOURCE uses BERTTopic to learn the topics represented in OTHER COLLECTION OF DOCUMENTS.
- ☐ Explain the characteristics of the ML models/techniques that were used in each of the references that your team has cited. For instance: At a high level, LDA works as follows: HIGH LEVEL EXPLANATION OF LDA. On the other hand, BERTTopic works as follows: HIGH LEVEL EXPLANATION OF BERTTopic. In general, the advantages of LDA are that ADVANTAGES OF LDA, whereas it has the disadvantages that DISADVANTAGES OF LDA. On the other hand, BERTTopic has the advantages that ADVANTAGES OF BERTTopic, whereas its disadvantages are that DISADVANTAGES OF BERTTopic.
- ☐ Include at least 1 reference to a dataset that will allow your team to build a similar ML system during the course. For instance: We could use the dataset DATASET mentioned in SOURCE to build a POC ML system for topic modeling. The data is available at LINK/WEBSITE. Alternatively, we could also use OTHER DATASET, mentioned in OTHER SOURCE. This other dataset can be downloaded at LINK/WEBSITE.
- ☐ Include a description of the data available in each of the datasets that your team has cited. For instance: The data in DATASET include NUMBER OF DOCUMENTS IN DATASET for a total of NUMBER OF WORDS IN DATASET. Additional information in DATASET includes ADDITIONAL INFORMATION IN DATASET. Similarly, the data in OTHER DATASET includes NUMBER OF DOCUMENTS IN OTHER

DATASET for a total of NUMBER OF WORDS IN OTHER DATASET. OTHER DATASET also includes ADDITIONAL INFORMATION IN OTHER DATASET.

Other notes

- Keep in mind that your team's goal in the course is to build a POC ML system. It is advisable to keep things simple - or at least manageable - when it comes to the choice of the data and of the models that your team will experiment with. A good rule of thumb: if your team can't build its POC ML system on a modern laptop - e.g., because the dataset your team selected is too large for it, because the model your team is trying to train/fit requires more computing power, ... - it likely means that the POC ML system your team is trying to build is too complex (for this course, anyway).
- For this project, it is advisable that your team chooses a business problem that can be addressed using an "online" ML system. Business problems that can be framed in terms of e.g., regression, classification, recommendation of contents, search and retrieval, ... tend to lend themselves better to online inference than business problems that can be instead framed in terms of e.g., clustering, dimensionality reduction, ... This is not a strict requirement for the project. It will just make your team's life a little easier in the next few weeks.
- Your team is free to manage the project's code and other assets in any way the team sees fit. However, it is advisable to make good use of a version control tool such as git. Consider creating a repository on GitHub for your team's work. If you have never used a version control tool in the past, this is a good opportunity to learn the basics.

Data Sources

1. **U.S. Census Bureau:** Offers comprehensive data on the U.S. population, economy, and more. Website: <https://www.census.gov> (<https://www.census.gov/>)
2. **CIA World Factbook:** Provides information on the history, people, government, economy, and geography of countries worldwide. Website: <https://www.cia.gov/the-world-factbook/> (<https://www.cia.gov/the-world-factbook/>)
3. **Data.gov:** The U.S. government's open data site, offering datasets across various domains. Website: <https://www.data.gov> (<https://www.data.gov/>)
4. **Dataverse Project:** An open-source repository for research data across disciplines. Website: <https://dataverse.org> (<https://dataverse.org/>)
5. **Gapminder:** Provides global statistics with tools for data visualization. Website: <https://www.gapminder.org> (<https://www.gapminder.org/>)
6. **Google Public Data Explorer:** Allows users to explore vast datasets on various topics. Website: <https://www.google.com/publicdata/directory> (<https://www.google.com/publicdata/directory>)
7. **HealthData.gov:** Offers datasets related to health topics from U.S. government agencies. Website: <https://healthdata.gov> (<https://healthdata.gov/>)

8. **Kaggle Datasets:** A platform with diverse datasets for data science and machine learning projects. Website: <https://www.kaggle.com/datasets> (<https://www.kaggle.com/datasets>)
9. **Linked Data:** A method of publishing structured data to interlink and make it more useful. Website: <https://linkeddata.org> (<https://linkeddata.org/>)
10. **The Guardian DataBlog:** Provides datasets and analyses on various topics. Website: <https://www.theguardian.com/data> (<https://www.theguardian.com/data>)
11. **NASA Climate Data:** Offers datasets related to climate change and environmental data. Website: <https://climate.nasa.gov> (<https://climate.nasa.gov/>)
12. **NYC OpenData:** Provides datasets from New York City agencies. Website: <https://opendata.cityofnewyork.us> (<https://opendata.cityofnewyork.us/>)
13. **World Health Organization (WHO) Data:** Offers global health-related datasets. Website: <https://www.who.int/data> (<https://www.who.int/data>)
14. **World Bank Open Data:** Provides free and open access to global development data. Website: <https://data.worldbank.org> (<https://data.worldbank.org/>)
15. **UCI Machine Learning Repository:** A collection of databases for empirical machine learning research. Website: <https://archive.ics.uci.edu/ml/index.php> (<https://archive.ics.uci.edu/ml/index.php>)
16. **DataHub.io:** A platform for discovering and sharing datasets. Website: <https://datahub.io> (<https://datahub.io/>)
17. **Maven Analytics Data Playground:** Offers free datasets for data analysis and visualization practice. Website: <https://mavenanalytics.io/data-playground> (<https://mavenanalytics.io/data-playground>)
18. **FiveThirtyEight Datasets:** Offers datasets used in FiveThirtyEight's data-driven articles, covering topics like politics, sports, and culture. Website: <https://data.fivethirtyeight.com> (<https://data.fivethirtyeight.com/>)
19. **BuzzFeed News Datasets:** Provides datasets used in BuzzFeed News investigations, encompassing diverse topics. Website: <https://github.com/BuzzFeedNews/> (<https://github.com/BuzzFeedNews/>)
20. **UCI Machine Learning Repository:** A collection of databases, domain theories, and datasets widely used for empirical and machine learning research. Website: <https://archive.ics.uci.edu/ml/index.php> (<https://archive.ics.uci.edu/ml/index.php>)
21. **DataHub.io:** A platform for discovering, sharing, and using high-quality data sets in various formats. Website: <https://datahub.io> (<https://datahub.io/>)

22. **Humanitarian Data Exchange (HDX):** An open platform for sharing data across crises and organizations, managed by the United Nations Office for the Coordination of Humanitarian Affairs. Website: <https://data.humdata.org> (<https://data.humdata.org/>)
23. **World Bank Open Data:** Provides free and open access to global development data, including economic indicators, health statistics, and population metrics. Website: <https://data.worldbank.org> (<https://data.worldbank.org/>)
24. **Google Dataset Search:** A search engine specifically for datasets across the web, helping users find datasets stored across various repositories. Website: <https://datasetsearch.research.google.com> (<https://datasetsearch.research.google.com/>)
25. **Data USA:** A comprehensive visualization of U.S. public data, providing insights into the economy, education, and health. Website: <https://datausa.io> (<https://datausa.io/>)
26. **Data.gov.uk:** The UK government's open data portal, offering datasets on various topics, including government spending, crime statistics, and transportation data. Website: <https://data.gov.uk> (<https://data.gov.uk/>)
27. **Data Commons:** An open knowledge repository that integrates data from various sources, including the U.S. Census, World Bank, and more, providing a unified view of diverse datasets. Website: <https://datacommons.org> (<https://datacommons.org/>)
28. **Stanford Open Policing Project:** Standardized police stop data are available to download by location. Website: <https://openpolicing.stanford.edu/data/> (<https://openpolicing.stanford.edu/data/>)
29. **FRED (Federal Reserve Economic Data)**  (<https://fred.stlouisfed.org>): Comprehensive U.S. economic data, including inflation, GDP, employment, and interest rates.
30. **OECD Data**  (<https://data.oecd.org>): Global datasets on economics, education, and public policy from the Organization for Economic Co-operation and Development.
31. **IMF Data Portal**  (<https://data.imf.org>): Access to international financial and macroeconomic indicators from the International Monetary Fund.
32. **NOAA Climate Data Online**  (<https://www.ncdc.noaa.gov/cdo-web/>): Historical climate, temperature, and weather data.
33. **USGS Earth Explorer**  (<https://earthexplorer.usgs.gov>): Satellite and geospatial data for land use, elevation, and environmental monitoring.
34. **EPA Open Data**  (<https://www.epa.gov/open>): Environmental data on air quality, water quality, and hazardous waste from the U.S. Environmental Protection Agency.
35. **Pew Research Center**  (<https://www.pewresearch.org/download-datasets>): Survey datasets on public opinion, social trends, and media use.

36. **Bureau of Labor Statistics (BLS)** [↗\(https://www.bls.gov/data\)](https://www.bls.gov/data): Employment, wage, and price data across U.S. industries.
37. **Common Crawl** [↗\(https://commoncrawl.org\)](https://commoncrawl.org): Web-scale datasets for text and NLP research, including billions of webpages.
38. **Google Cloud Public Datasets** [↗\(https://cloud.google.com/public-datasets\)](https://cloud.google.com/public-datasets): Hosted datasets for machine learning, public health, and climate science.
39. **OpenStreetMap** [↗\(https://www.openstreetmap.org\)](https://www.openstreetmap.org): Free and editable global geographic data.
40. **WorldPop** [↗\(https://www.worldpop.org\)](https://www.worldpop.org): Population distribution and demographic data with global coverage.
41. **OpenFlights** [↗\(https://openflights.org/data.html\)](https://openflights.org/data.html): Comprehensive data on airports, airlines, and flight routes.
42. **Spotify Developer API** [↗\(https://developer.spotify.com/documentation/web-api\)](https://developer.spotify.com/documentation/web-api): Access to track, artist, and playlist data for music analysis.
43. **The Movie Database (TMDB) API** [↗\(https://developer.themoviedb.org\)](https://developer.themoviedb.org): Film and TV metadata for entertainment analytics.
44. **Global Forest Watch** [↗\(https://data.globalforestwatch.org\)](https://data.globalforestwatch.org): Forestation and land use data for sustainability studies.
45. **CDC WONDER** [↗\(https://wonder.cdc.gov\)](https://wonder.cdc.gov): U.S. public health datasets including mortality, births, and disease prevalence.
46. **IPUMS** [↗\(https://ipums.org\)](https://ipums.org): Census and survey microdata from the U.S. and around the world for demographic analysis.
47. **Esri Open Data Hub** [↗\(https://hub.arcgis.com\)](https://hub.arcgis.com): Geospatial and mapping datasets from local governments and organizations.
48. **FRED Graph Data API** [↗\(https://fred.stlouisfed.org/docs/api/fred/\)](https://fred.stlouisfed.org/docs/api/fred/): API access for downloading and visualizing Federal Reserve datasets.
49. **Open Data Network** [↗\(https://www.opendatanetwork.com\)](https://www.opendatanetwork.com): Aggregated public datasets on finance, health, and infrastructure.
50. **Global Biodiversity Information Facility (GBIF)** [↗\(https://www.gbif.org\)](https://www.gbif.org): Worldwide biodiversity and species occurrence data.

Project Proposal Rubric

Criteria	Ratings			Pts
Project Proposal Rubric	3 pts Complete and meets expectations The proposal clearly describes a business problem and explains why it matters from a business perspective. It includes at least one source where the problem has been solved with ML, with a high-level explanation of the techniques used (including strengths and weaknesses). It identifies at least one relevant dataset with a description of what the data contains and where to access it. The proposal is well-organized and within the page limit.	2 pts Partially complete The proposal addresses the business problem but is missing or underdeveloped in one or two areas. For example: sources are cited but the ML techniques are not explained, a dataset is identified but not described, the business justification is vague, or the connection between the problem, the existing solutions, and the chosen data is unclear.	1 pts Incomplete The proposal is missing most of the required components. For example: no source with an existing ML solution is provided, no dataset is identified, or the business problem is not clearly defined. The submission shows minimal effort or does not provide enough detail for the team to move forward with the project.	3 pts
Total Points: 3				